



**Roll No:**

**MCA-INT**  
**(SEM V) THEORY EXAMINATION 2024-25**  
**SYSTEM ANALYSIS & DESIGN**

**TIME: 3 HRS**

**M.MARKS: 70**

**Note:** Attempt all Sections. In case of any missing data; choose suitably.

## SECTION A

1. Attempt all questions in brief.

$$2 \times 07 = 14$$

| Q no. | Question                                          | CO | Level |
|-------|---------------------------------------------------|----|-------|
| a.    | Explain open system and closed system.            | 1  | 2     |
| b.    | How Information System is useful to us?           | 1  | 2     |
| c.    | List out the subcategories of Information System. | 1  | 2     |
| d.    | What is the role of feasibility study.            | 2  | 2     |
| e.    | Explain the term onsite observations.             | 3  | 2     |
| f.    | Explain the process for computer contract.        | 5  | 2     |
| g.    | What do you mean by behavioral feasibility study. | 4  | 2     |

## SECTION B

2. Attempt any *three* of the following:

**07 x 3 = 21**

|    |                                                                                                                            |   |   |
|----|----------------------------------------------------------------------------------------------------------------------------|---|---|
| a. | What do you mean by Structured Analysis? Briefly explain the tools used for structured Analysis.                           | 1 | 2 |
| b. | What is the use of Data Flow Diagram? What are the symbols used in DFD? Draw the DFD for library management system.        | 2 | 3 |
| c. | Explain the term Software Development Life Cycle Model. With a neat diagram explain the various phases of Waterfall Model. | 3 | 2 |
| d. | Discuss in detail the Interview Technique for information gathering.                                                       | 4 | 2 |
| e. | What is the procedure for H/W & S/W selection?                                                                             | 5 | 2 |

## SECTION C

3. Attempt any one part of the following:

**07 x 1 = 07**

|    |                                                                        |   |   |
|----|------------------------------------------------------------------------|---|---|
| a. | Define a system. Discuss the characteristics and elements of a system. | 1 | 2 |
| b. | How MIS is different from DSS?                                         | 1 | 2 |

4. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

|    |                                                                       |   |   |
|----|-----------------------------------------------------------------------|---|---|
| a. | What is SRS? List out the different types of requirements put in SRS? | 2 | 2 |
| b. | Discuss the various types of maintenance.                             | 2 | 2 |

5. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

|    |                                                                                        |   |   |
|----|----------------------------------------------------------------------------------------|---|---|
| a. | Explain in detail the interview and questionnaire technique for information gathering. | 4 | 2 |
| b. | Explain the term Decision Table, Structured English.                                   | 4 | 2 |

6. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

|    |                                                                                               |   |   |
|----|-----------------------------------------------------------------------------------------------|---|---|
| a. | "Review of written documents is a good method of information gathering". Justify your answer. | 3 | 2 |
| b. | Discuss the behavioral issues involved in understanding the analyst/user interface.           | 3 | 2 |

7. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

|    |                                                                                                                              |   |   |
|----|------------------------------------------------------------------------------------------------------------------------------|---|---|
| a. | Discuss the issues for the computer contract.                                                                                | 5 | 2 |
| b. | Why is Feasibility Study conducted? What are the various feasibility studies carried out during the development of software? | 3 | 2 |